25901202 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCFS201 - APPLIED CRYPTOGRAPHY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Define brute-force attack. [U] / [CO1]
2. Write the number of rounds in AES-128, AES-192, and AES-256. [C] / [CO4]
3. State the need for digital certificates. [A] / [CO3]
4. Define substitution cipher. [U] / [CO1]
5. Define Stream cipher. [An] / [CO2]
6. Name the two main phases of sponge construction. [C] / [CO4]
7. Discuss cryptography ensure security in online banking. [A] / [CO3]
8. Identify the main security risk in symmetric encryption. [An] / [CO2]
9. List the public parameters required in the Diffie–Hellman algorithm. [A] / [CO3]

10. Write which cryptographic primitive is used to ensure data confidentiality. [An] / [CO2]

11. Define ECB. [C] / [CO4]

12. Define a key exchange protocol. [A] / [CO3]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain is the use of hash functions in digital signatures. [C] / [CO4]

14. Explain DES operation in detail. [An] / [CO2]

15. Write the steps of key generation in RSA Algorithm. [A] / [CO3]

16. Explain the concept of asymmetric key cryptography. [An] / [CO2]

17. Write the advantages of CBC over ECB. [C] / [CO4]

18. Explain the steps involved in generating and verifying an HMAC. [An] / [CO2]

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19. Illustrate the concept of symmetric key cryptography in detail. [An] / [CO2]

20. Explain different block cipher modes of operation. [C] / [CO4]

21. Explain Public Key Infrastructure (PKI) with its objectives, architecture, and major components. [A] / [CO3]

[10x2 = 20]